

HUNDRED DAYS OF CODE (DAY – 7)

❖ Question 1:

Write a C program to find acceleration due to gravity at height when Earth radius, height, and surface gravity are given.

- We must create a C program that can calculate acceleration due to gravity at height when Earth's radius, height, and surface gravity are given.

Basically, using radius, height and surface gravity as input and calculating acceleration due to gravity as output.

So, we will use the following formula:

$$g_h = g \left(\frac{R}{R + h} \right)^2$$

Where:

g_h = acceleration due to gravity at height h

g = acceleration due to gravity at Earth's surface

R = radius of Earth

h = height above Earth's surface

• Input / Output Table:

Test Case	Input	Expected Output	Actual Output	Result
1	h = 150	gh = 9.36	gh = 9.36	Pass
2	h = 800	gh = 7.74	gh = 7.74	Pass
3	h = 1210	gh = 6.93	gh = 6.93	Pass

- **Algorithm:**

- 1) Start
- 2) Declare variables R , h , g , and g_h .
- 3) Read the Earth's radius (R).
- 4) Read the height above Earth's surface (h).
- 5) Read the surface gravity (g).
- 6) Calculate acceleration due to gravity at height using:
$$g_h = g \times (R / (R + h))^2$$
- 7) Display the value of g_h .
- 8) Stop.

- **C Program:**

```
#include <stdio.h>
int main()
{
    float h, gh;
    float R = 6400;
    float g = 9.8;
    printf("Enter height above Earth's surface: ");
    scanf("%f", &h); gh = g * (R / (R + h)) * (R / (R + h));
    printf("Acceleration due to gravity = %.2f m/s^2", gh);
    return 0;
}
```

Note: I took help from chatgpt on using m/s^2 on the 8th line.

- **Compile it with:**

```
gcc hdoc7a.c -o hdoc7a -lm
```

- **Run it with:**

```
./hdoc7a
```

```
Session Actions Edit View Help

(kali㉿kali)-[~/Desktop/HDOC]
$ vi hdoc7a.c

(kali㉿kali)-[~/Desktop/HDOC]
$ cc hdoc7a.c

(kali㉿kali)-[~/Desktop/HDOC]
$ gcc hdoc7a.c -o hdoc7a -lm
```

Test Cases:

```
(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc7a
Enter height above Earth's surface: 150
Acceleration due to gravity = 9.36 m/s^2

(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc7a
Enter height above Earth's surface: 800
Acceleration due to gravity = 7.74 m/s^2

(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc7a
Enter height above Earth's surface: 1210
Acceleration due to gravity = 6.93 m/s^2

(kali㉿kali)-[~/Desktop/HDOC]
$
```

❖ QUESTION 2:

Write a C program to find electric flux when electric field, area, and angle are given. We must create a C program in which we take electric field, area and angle as input and calculate electric flux as output.

We can do that by using the following formula:

$$\Phi = EA \cos \theta$$

Where:

Φ = electric flux

E = electric field

A = area of the surface

θ = angle between the electric field and the normal (perpendicular) to the surface

- **INPUT / OUTPUT TABLE:**

Test Case	Input	Expected Output	Actual Output	Result
1	Electric field = 100 Area = 3 Angle = 45	Electric flux = 212.13	Electric flux =212.13	Pass
2	Electric field = 80 Area = 4 Angle = 30	Electric flux =277.13	Electric flux = 277.13	Pass
3	Electric field = 80 Area = 2 Angle = 60	Electric flux =80.00	Electric flux = 80.00	Pass

- **ALGORITHM:**

- 1) Start
- 2) Declare variables E, A, angle, and flux.
- 3) Read the electric field (E).
- 4) Read the area (A).
- 5) Read the angle (θ) in degrees.
- 6) Convert the angle from degrees to radians.
- 7) Calculate electric flux using:
$$\text{flux} = E \times A \times \cos(\text{angle})$$
- 8) Display the value of electric flux.
- 9) Stop.

- **C PROGRAM:**

```
#include <stdio.h>
#include <math.h>
int main()
{
    float E;
    float flux;
    printf("Enter electric field: ");
    scanf("%f", &E);
    float A;
```

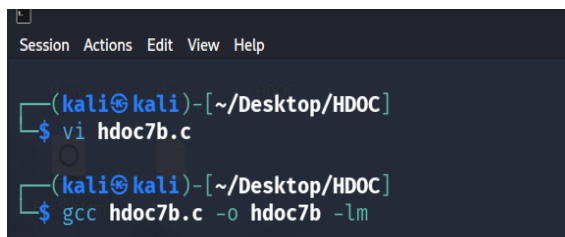
```
printf("Enter area: ");
scanf("%f", &A);
float angle;
printf("Enter angle in degrees: ");
scanf("%f", &angle);
flux = E * A * cos(angle * 3.14 / 180);
printf("Electric flux = %.2f", flux);
return 0;
}
```

Compile it with:

```
gcc hdoc7b.c -o hdoc7b -lm
```

Run it with:

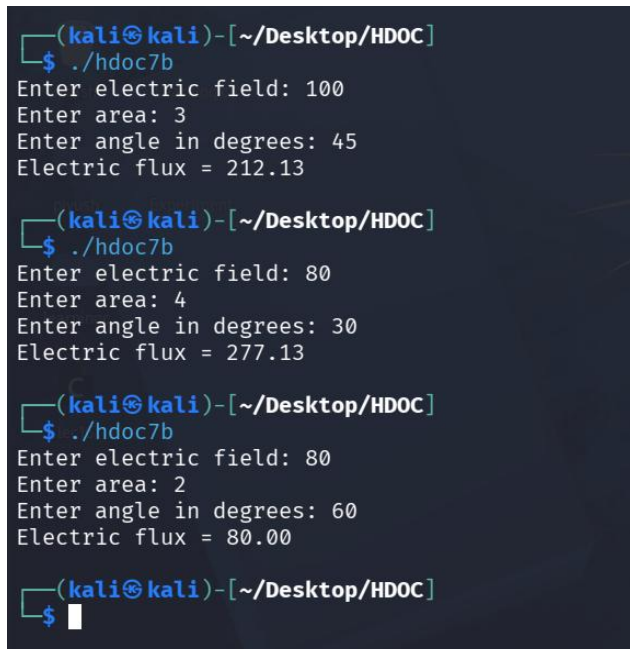
```
./hdoc7b
```

A terminal window with a dark background and light blue text. The window title is "Session Actions Edit View Help". The prompt is "(kali㉿kali)-[~/Desktop/HDOC]". The user enters "vi hdoc7b.c" and then "gcc hdoc7b.c -o hdoc7b -lm".

```
(kali㉿kali)-[~/Desktop/HDOC]
$ vi hdoc7b.c

(kali㉿kali)-[~/Desktop/HDOC]
$ gcc hdoc7b.c -o hdoc7b -lm
```

Test Cases:

A terminal window with a dark background and light blue text. The window title is "Session Actions Edit View Help". The prompt is "(kali㉿kali)-[~/Desktop/HDOC]". The user enters " ./hdoc7b" three times, each time followed by input for electric field, area, and angle, and then the resulting electric flux.

```
(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc7b
Enter electric field: 100
Enter area: 3
Enter angle in degrees: 45
Electric flux = 212.13

(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc7b
Enter electric field: 80
Enter area: 4
Enter angle in degrees: 30
Electric flux = 277.13

(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc7b
Enter electric field: 80
Enter area: 2
Enter angle in degrees: 60
Electric flux = 80.00

(kali㉿kali)-[~/Desktop/HDOC]
$
```

❖ QUESTION 3:

Write a C program to find relative error when absolute error and measured value are given.

We must create a C program in which we take absolute error and measure value as input and calculate relative error as output.

We need to use the following formula:

Relative Error = Absolute error/ Measured value

• INPUT / OUTPUT TABLE:

Test Case	Input	Expected Output	Actual Output	Result
1	Absolute error = 0.5 Measured value = 5	Relative error= 0.1000	Relative error=0.1000	Pass
2	Absolute error = 0.8 Measured value =10	Relative error=0.0800	Relative error=0.0800	Pass
3	Absolute error = 0.225 Measured value =15	Relative error=0.0167	Relative error=0.0167	Pass

• ALGORITHM:

- 1) Start
- 2) Declare variables absolute error, measured value, and relative error.
- 3) Enter the absolute error.
- 4) Enter the measured value.
- 5) Calculate relative error using:
relative error = absolute error / measured value
- 6) Display the relative error.
- 7) Stop.

• C PROGRAM:

```
#include<stdio.h>

int main()
{
float absolute_error, measured_value, relative_error;

printf("Enter absolute error: ");

scanf("%f", &absolute_error);

printf("Enter measured value: ");

scanf("%f", &measured_value);

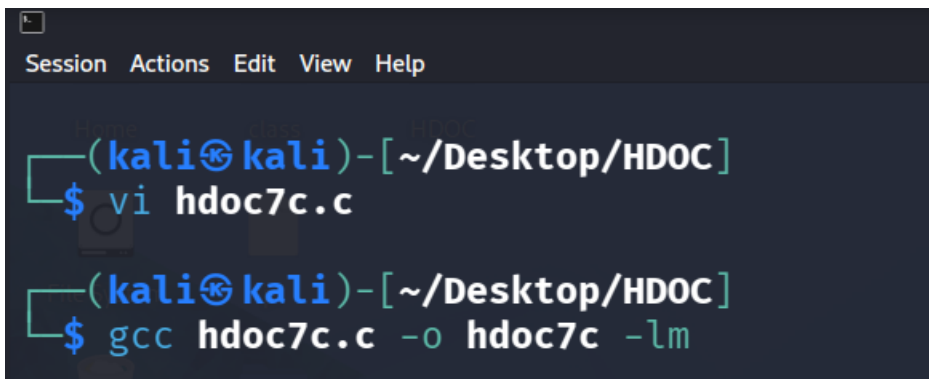
relative_error = absolute_error / measured_value;

printf("Relative error = %.4f", relative_error);

return 0;
}
```

Compile it with:

```
gcc hdoc7c.c -o hdoc7c -lm
```

A screenshot of a terminal window with a dark background. The window has a title bar with a small icon and the text "Session Actions Edit View Help". The terminal shows two prompts. The first prompt is "(kali㉿kali)-[~/Desktop/HDOC]" followed by "\$ vi hdoc7c.c". The second prompt is "(kali㉿kali)-[~/Desktop/HDOC]" followed by "\$ gcc hdoc7c.c -o hdoc7c -lm".

```
(kali㉿kali)-[~/Desktop/HDOC]
$ vi hdoc7c.c

(kali㉿kali)-[~/Desktop/HDOC]
$ gcc hdoc7c.c -o hdoc7c -lm
```

Run it with:

```
./hdoc7c
```

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ ./hdoc7c
```

Enter absolute error:0.5

Enter measured value5

Relative error = 0.1000

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ ./hdoc7c
```

Enter absolute error:0.8

Enter measured value10

Relative error = 0.0800

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ ./hdoc7c
```

Enter absolute error:0.25

Enter measured value15

Relative error = 0.0167

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ █
```